TASK 1 — BASIC NETWORK SNIFFER

Objective

Build a Python program that captures network traffic packets and displays useful information such as source IP, destination IP, protocol, and payload. This follows the CodeAlpha task requirement to use libraries such as Scapy or socket for packet capturing and to study packet structure and network flow.

Tools and Technologies

• Python 3
• Scapy
• Command line / terminal
• Authorized local machine or lab environment

Implementation

The following program uses Scapy to sniff packets and print basic packet information. It is intended for an authorized environment only.

```
from scapy.all import sniff, IP, TCP, UDP, ICMP, Raw

def process_packet(packet):
if IP in packet:
src = packet[IP].src
dst = packet[IP].dst

if TCP in packet:
protocol = "TCP"
elif UDP in packet:
protocol = "UDP"
elif ICMP in packet:
protocol = "ICMP"
else:
protocol = "Other"

payload = ""
if Raw in packet:
payload = bytes(packet[Raw].load)[:40]

print(f"Source : {src}")
print(f"Destination : {dst}")
print(f"Protocol : {protocol}")
print(f"Payload : {payload}")
print("-" * 50)

print("Starting packet capture...")
sniff(prn=process_packet, count=20, store=False)
```

Working Procedure

1. Install Scapy using the Python package manager.
2. Run the program with the required permissions in an authorized lab environment.
3. The sniffer captures a limited number of packets.
4. For IP packets, the source and destination addresses are extracted.
5. The program identifies common protocols such as TCP, UDP, and ICMP.
6. A short payload preview is displayed when available.
7. The observed information can be used to understand how packets move through a network.

Expected Output

The terminal displays packet records containing source IP, destination IP, protocol, and an optional payload preview. The exact output depends on the traffic present during the authorized capture.

Result

The task demonstrates basic packet capture and analysis. It provides practical understanding of network traffic structure and common protocols while emphasizing authorized and responsible security testing.

Safety Note

Packet capture should be performed only on systems and networks for which permission has been provided. The CodeAlpha instructions emphasize completing projects and submitting the completed work, while the internship report also states that security testing should be controlled and authorized.